Phy 211

Lectures Notes

By

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Lecture 2

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➢ Coulomb's law (inverse square law):

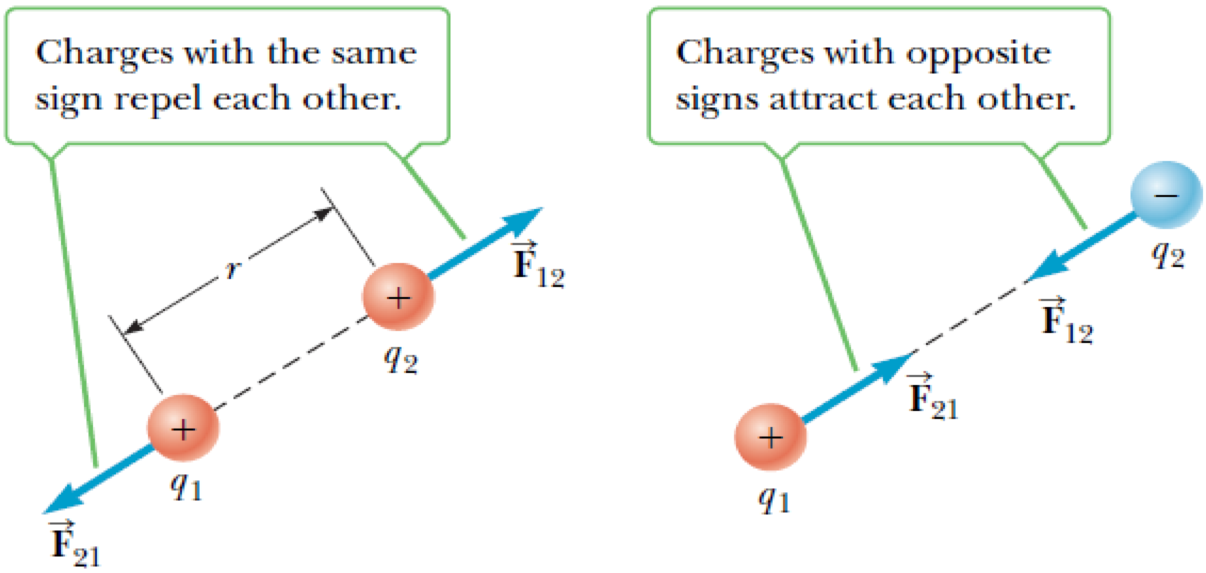
- The magnitude of the electric force between two **point** charges q_1 & q_2 that are separated by distance r is given by:

$$\boxed{F = K\frac{|q_1||q_2|}{r^2}}$$

$$K: Coulomb\ constant \rightarrow K = 9 \times 10^9\ \frac{\mathrm{N \cdot m^2}}{\mathrm{C^2}}$$

$$K = \frac{1}{4\pi\varepsilon_o}$$

ε_o: Permittivity of free space ($8.85 \times 10^{-12}\ \frac{\mathrm{C^2}}{\mathrm{N \cdot m^2}}$)



$F_{12}: force\ acting\ by\ q_1\ on\ q_2$

$F_{21}: force\ acting\ by\ q_2\ on\ q_1$

$$\boxed{F_{12} = F_{21}\ \ (equal\ magnitude) \qquad \overrightarrow{F_{12}} = -\overrightarrow{F_{21}}\ \ (opposite\ direction)}$$

- **The force on q_1 is equal in magnitude and opposite in direction to the force on q_2 ($\overrightarrow{F_{12}} = -\overrightarrow{F_{21}}$).**
- **The electric force is a vector quantity (has magnitude and direction).**

- **Example 4:**

The electron and proton of a hydrogen atom are separated (on the average) by a distance of about $5.3 \times 10^{-11}\ m$. Find the magnitudes of the electric force and the gravitational force that each particle exerts on the other, and the ratio of the electric force F_e to the gravitational force F_g.

$(m_e = 9.1 \times 10^{-31}\text{kg}, m_p = 1.67 \times 10^{-27}\ \text{kg} \ \&\ G = 6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2)$

Solution

$$F_e = K\frac{|q_1||q_2|}{r^2} = K\frac{|e|^2}{r^2} = 9 \times 10^9 \frac{(1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2} = 8.2 \times 10^{-8}\ \text{N}$$

$$F_g = G\frac{m_1 m_2}{r^2} = G\frac{m_e m_p}{r^2} = 6.67 \times 10^{-11}\frac{(9.1 \times 10^{-31})(1.67 \times 10^{-27})}{(5.3 \times 10^{-11})^2}$$
$$= 3.6 \times 10^{-47}\ \text{N}$$

$$\frac{F_e}{F_g} = \frac{8.2 \times 10^{-8}}{3.6 \times 10^{-47}} = 2.3 \times 10^{39}$$

$$\because F_e \gg F_g$$

The electric force is predominant is the microscopic scale

- **Q1:** If the distance between two charges is doubled, by what factor is the magnitude of the electric force changed?

Solution

$$\boldsymbol{F_1 = K\frac{q_1 q_2}{r^2}}$$

$$\boldsymbol{F_2 = K\frac{q_1 q_2}{(2r)^2} = \frac{1}{4}K\frac{q_1 q_2}{r^2} = \frac{F_1}{4}}$$

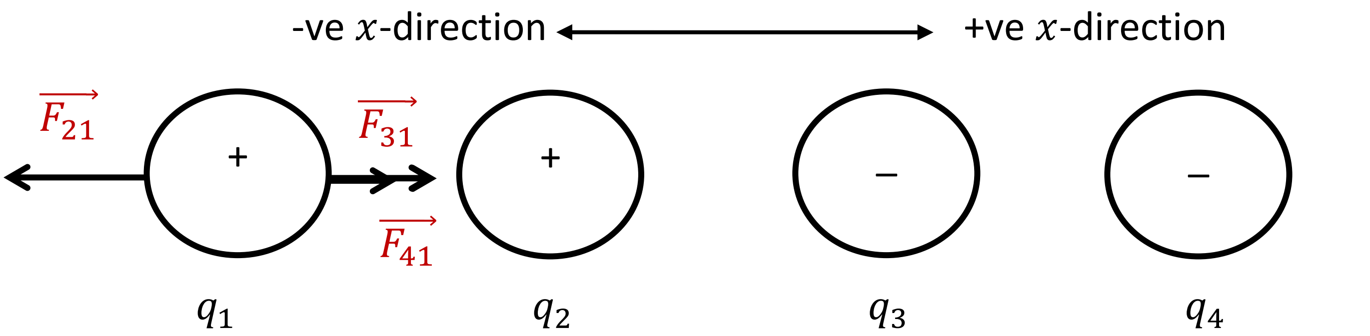
The electric force is a changed by a factor of $\frac{1}{4}$.

➢ Several charges in a line:

- The net force on a charge is the vector sum of all forces acting on it.

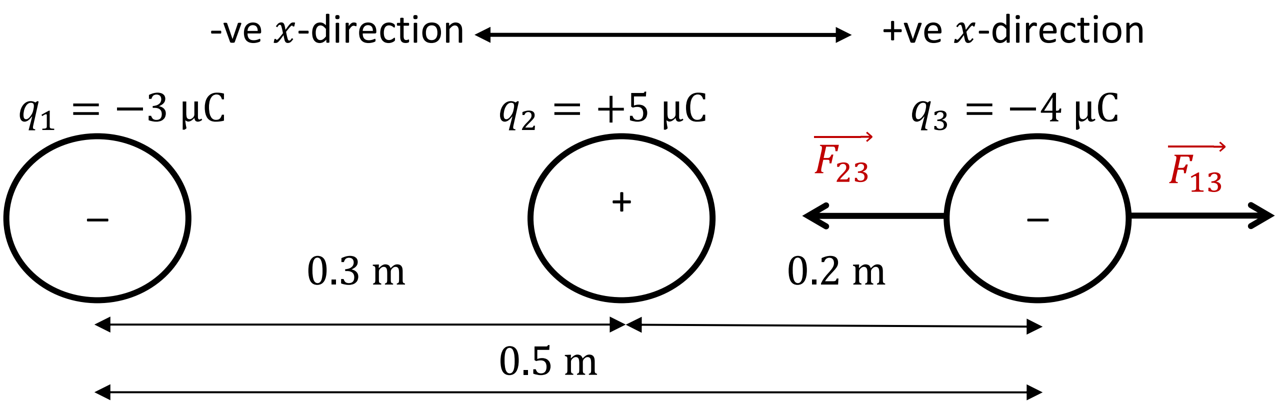
$$\overrightarrow{F_1} = \overrightarrow{F_{21}} + \overrightarrow{F_{31}} + \overrightarrow{F_{41}} + \overrightarrow{F_{51}} + \cdots$$

- We set the positive direction and any force in the opposite direction is given a –ve sign



$$\overrightarrow{F_1} = \overrightarrow{F_{21}} + \overrightarrow{F_{31}} + \overrightarrow{F_{41}} = -F_{21} + F_{31} + F_{41}$$

- **Example 5:**



Calculate the net force on the charge q_3 due to the other charges q_1 & q_2

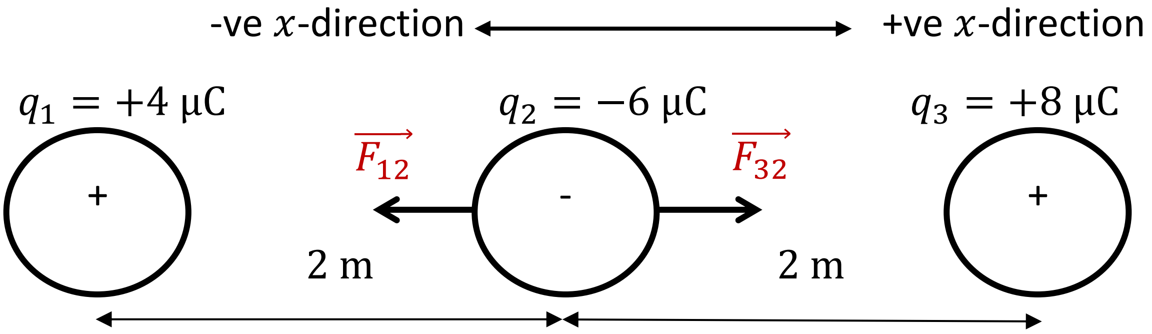
Solution

$$F_{13} = K\frac{|q_1||q_3|}{r_{31}^2} = 9\times 10^9 \frac{(3\times 10^{-6})\times(4\times 10^{-6})}{0.5^2}$$
$$= 0.432\ \mathrm{N}\ (in + ve\ x\ direction)$$

$$F_{23} = K\frac{|q_2||q_3|}{r_{32}^2} = 9\times 10^9 \frac{(5\times 10^{-6})\times(4\times 10^{-6})}{0.2^2}$$
$$= 4.5\ \mathrm{N}\ (in - ve\ x\ direction)$$

$$\overrightarrow{F_3} = F_{13} - F_{23} = 0.432 - 4.5 = -4.068\ N \rightarrow \therefore \overrightarrow{F_3} = 4.068\ \mathrm{N}\ \ in\ - ve\ x\ direction$$

- **Example 6:**



Calculate the net force on the charge q_2 due to the other charges q_1 & q_3

<u>**Solution**</u>

$$F_{12} = K\frac{|q_1||q_2|}{r_{12}^2} = 9 \times 10^9 \frac{(4 \times 10^{-6}) \times (6 \times 10^{-6})}{2^2}$$
$$= 0.054\ \text{N}\ (in - ve\ x\ direction)$$

$$F_{32} = K\frac{|q_3||q_2|}{r_{32}^2} = 9 \times 10^9 \frac{(8 \times 10^{-6}) \times (6 \times 10^{-6})}{2^2}$$
$$= 0.108\ (in + ve\ x\ direction)$$

$$\overrightarrow{F_2} = F_{32} - F_{12} = 0.108 - 0.054 = +0.054\ \text{N}$$

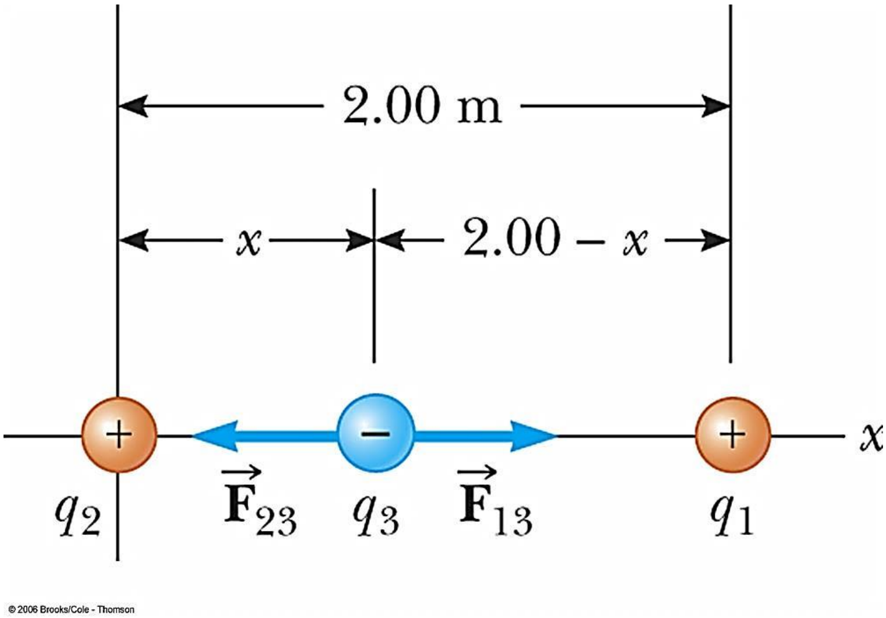
$\overrightarrow{F_2} = 0.054\ N$ (in +ve x direction) (in direction of F_{32})

1- Calculate the net force on the charge q_1 due to the other charges q_2 & q_3

2- Calculate the net force on the charge q_3 due to the other charges q_1 & q_2

- **Problem:**

Three charges lie along the x-axis as in Figure. The positive charge $q_1 = 4\ \mu\text{C}$ is at $x = 2.0\ \text{m}$, and the positive charge $q_2 = 8\ \mu\text{C}$ is at the origin. Where must a negative charge q_3 be placed on the x-axis so that the resultant electric force on it is zero?



Solution

q_3 must be placed between q_1 & q_2 So that $\overrightarrow{F_{23}}$ & $\overrightarrow{F_{13}}$ be in opposite direction.

Let q_3 be place at distance x from q_2 at the origin

$$F_3 = F_{13} - F_{23} = 0$$

$$K\frac{|q_1||q_3|}{r_{13}^2} - K\frac{|q_2||q_3|}{r_{23}^2} = 0$$

$$\frac{|q_1|}{(2-x)^2} - \frac{|q_2|}{x^2} = 0 \longrightarrow \frac{|q_1|}{(2-x)^2} = \frac{|q_2|}{x^2}$$

$$\frac{4}{(2-x)^2} = \frac{8}{x^2} \longrightarrow 4x^2 = 8(2-x)^2$$

$$4x^2 = 8(4 - 4x + x^2) \longrightarrow 4x^2 = 32 - 32x + 8x^2$$

$$4x^2 - 32x + 32 = 0 \longrightarrow x^2 - 8x + 8 = 0$$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{8 \pm \sqrt{64 - 32}}{2} = \frac{8 \pm \sqrt{32}}{2}$$

$x = 6.82\ is\ not\ acceptable$ $\quad \therefore \boldsymbol{x = 1.171\ \text{m}}$